

problem and the problem of time-varying volatility, which also causes regular ordinary least squares to overreject the null of predictability too often.²³ Table 8.7 considers a variety of variables used in the literature ranging from macro variables, interest rates and spreads, and past excess returns.

Table 8.7 shows that it is very hard to predict returns. I highlight in bold in Table 8.7 all correlation coefficients that are significant at the 95% level—that is, they have p -values smaller than 0.05 (or equivalently they have t -statistics greater than 1.96 in absolute value). There are only two variables that have statistically significant correlations at the 95% level: Shiller’s (2000) cyclically adjusted ten-year earnings yield and the deviations from long-run consumption-wealth trends constructed by Lettau and Ludvigson (2001a) known as “cay.” The latter has look-ahead bias (it is constructed using information not available to investors at the time), so should be interpreted with caution because it cannot be used in an investable strategy.²⁴ The dividend yield has borderline predictive ability at the five-year horizon, with a t -statistic of 1.95. Thus, it is very hard to predict returns, just as Ross and Zhou say it should be.

The Shiller inverse price–earnings ratio that does so well in Table 8.7 and the borderline dividend yield at the five-year horizon is consistent with the Gordon model intuition in equation (8.4). High yields mean low prices. Low prices result from future cash flows being discounted at high expected returns. Consequently, the low price today forecasts that future returns tend to be high. Note that the only significant variables in Table 8.7 are these *valuation* ratios. Other predictors are documented in an extensive academic literature over different time horizons and samples.²⁵ Overall, it is hard to predict returns.

Be Wary of Long-Horizon R^2 s

Many commentators argue that there is substantial evidence of long-horizon predictability in data. Table 8.7 does not show this. It is there, but, again, it is hard to detect. You cannot conduct statistical inference at long horizons by using standard R^2 measures. The overlapping observations in equation (8.7) cause long-horizon R^2 s to be grossly inflated; the simple R^2 s appear to indicate substantial predictability, but it isn’t actually there. For example, take the dividend yield correlation of 49% with excess returns over the next five years. This implies an $R^2 = (0.49)^2 = 24\%$. This looks enormous, especially given that financial theory predicts that the regression R^2 s should be close to zero.

²³ The t -statistics in Table 8.7 are computed using Hodrick (1992) standard errors, which Ang and Bekaert (2007) show have the correct statistical properties and do not overreject the null of predictability. There are many versions of robust standard errors that still are not “robust” enough, including the commonly used Hansen and Hodrick (1980) and Newey and West (1987) standard errors, in the sense that they do not have correct size properties.

²⁴ See Brennan and Xia (2005).

²⁵ See a summary article by Rapach and Zhou (2011), who show that combinations of variables can do a better job than individual variables at forecasting equity returns.